THE AVERAGE FIRST LETTER OF DERANGEMENTS WITH ONE DESCENT HAS DENOMINATOR 2^{n-2}

MATH0

Abstract. For $n \geq 2$ let $T(n,1)$ be the denominator, in lowest terms, of the average first letter of the derangements of $\{1,\dots,n\}$ with exactly one descent; these numbers form the first column of the triangle A332010 of the OEIS, contributed by Peter Kagey in 2020. We prove Kagey's conjecture, recorded in that entry, that $T(n,1) = 2^{n-2}$: the derangements in question are exactly the 2^{n-2} words $s\,n\,1\,\bar{s}$, where s is the increasing word on a subset of $\{2,\dots,n-1\}$ and $\bar{s}$ the increasing word on its complement, and their first letters sum to $3\cdot 2^{n-2}-1$, which is coprime to 2^{n-2} (it is odd for $n \geq 3$). To our knowledge this is the first proof of the conjecture; the argument is elementary and also gives the companion conjecture for the numerators in A332009. The theorem is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry (not part of the formal-conjectures repository), whose definitions are kernel-checked against the first 25 listed terms.

1. Introduction

For $n \geq 1$ we write a permutation of $[n] = \{1,\dots,n\}$ in one-line notation, as the word $w = w_1 w_2 \cdots w_n$ of its values. It is a *derangement* if $w_i \neq i$ for every i, and $i \in \{1,\dots,n-1\}$ is a *descent* of w if $w_i > w_{i+1}$; let $\mathrm{des}(w)$ be the number of descents. The entry A332010 of the OEIS [13], contributed by Peter Kagey on February 4, 2020, is the irregular triangle (offset 2)

$$T(n,k) = \text{denominator, in lowest terms, of } \frac{1}{|\mathcal{D}_{n,k}|}\sum_{w\in\mathcal{D}_{n,k}} w_1,$$

where $\mathcal{D}_{n,k}$ is the set of derangements w of $[n]$ with $\mathrm{des}(w) = k$; row $n \geq 2$ consists of $T(n,1),\dots,T(n,n-1)$ for even n and $T(n,1),\dots,T(n,n-2)$ for odd n. The numerators form the companion entry A332009 [12]. For example, $\mathcal{D}_{4,1} = \{2341, 2413, 3412, 4123\}$ and the average first letter is $11/4$, so $T(4,1) = 4$.

The comments of A332010 contain the conjecture $T(n,1) = 2^{n-2}$, and those of A332009 the conjecture that the corresponding numerator is $A083329(n-1)$, where $A083329(j) = 3\cdot 2^{j-1}-1$ for $j \geq 1$ [10]. We prove both (Theorem 1 and Corollary 2); only the first is formalized. The idea is simple: a word with distinct letters and one descent is the concatenation of two increasing runs, and in a derangement the letter 1 cannot lie in the first run (it would sit at position 1) and the letter n cannot lie in the second run (it would sit at position n).

Prior work and status. The analogous denominators for all permutations are all 1, as the entry remarks: the expected first letter of a permutation of $[n]$ with k descents is $k+1$, a result of Conger [2]. Kagey's later paper [7] studies the expected value of the letters of permutations with a given number of k-cycles and does not treat derangements by descents. The number of derangements of $[n]$ with k descents is the triangle A219836 of David Callan [11]; its recurrence gives $|\mathcal{D}_{n,1}| = 2^{n-2}$, which is the denominator *before* reduction, and the entry links related work of Fu, Lin and Zeng [5] and of Alexandersson [1]. Derangements with prescribed descent sets

Date: October 19, 2026.

were enumerated by Han and Xin [6] and by Eriksen, Freij and Wästlund [3]; we found no treatment of the first letter there.

Both conjectures are still marked "Conjecture" in the current revisions of A332010 (#17) and A332009 (#18), dated March 30, 2024, in the copy of the OEIS database used by math0 (October 17, 2026), where no other entry refers to either sequence. The math0 referees' searches (arXiv, Mathematics Stack Exchange, the SeqFan list, and the papers above) found no proof; MathOverflow could not be searched. We therefore classify the result as new. The argument is short and elementary, and we do not claim that the description of one-descent derangements (Propositions 8–9) is new; what we did not find is any proof of the conjecture.

2. Statement

Theorem 1. *Let $n \geq 2$. Then* (1) $|\mathcal{D}_{n,1}| = 2^{n-2}$; (2) $\sum_{w \in \mathcal{D}_{n,1}} w_1 = 3 \cdot 2^{n-2} - 1$; *and* (3) *the average first letter of the elements of $\mathcal{D}_{n,1}$ is $(3 \cdot 2^{n-2} - 1)/2^{n-2} = 3 - 2^{2-n}$, a fraction in lowest terms; in particular $T(n, 1) = 2^{n-2}$.*

The formal statement is the last assertion, $T(n, 1) = 2^{n-2}$; parts (1) and (2) are proved in Lean as intermediate steps.

Corollary 2 (A332009; not formalized)**.** *For $n \geq 2$ the numerator of the average first letter of $\mathcal{D}_{n,1}$ in lowest terms is $3 \cdot 2^{n-2} - 1 = A083329(n-1)$.*

The name of A332010 [13] reads, verbatim, "`Irregular triangle of denominators of the average value of the first letter over all derangements of {1, 2, ..., n} with k descents.`", and its comments contain, verbatim, "`Even-indexed rows have length n - 1; odd-indexed rows have length n - 2.`" and

```
Conjecture:  T(n, 1) = 2^(n-2).
```

The comment is unsigned; the revision history of the entry shows that it was part of Peter Kagey's original submission of February 4, 2020. It gives no range of n; we read it for every row $n \geq 2$ of the triangle. The companion comment in A332009 reads "`Conjecture:  T(n, 1) = A083329(n - 1).`"

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A332010.lean`, reviewed for fidelity and frozen (SHA-256 `4e58e0c3...7cd`) before any proof attempt. Its definitions and the statement read, verbatim (omitting the import, comments and docstrings, blank lines, the theorems, and the auxiliary definitions `derB`, `desB`, `DC` and `rowC` used only for the kernel data check described below):

```
namespace Math0.Statements.A332010
def IsDerangement (w : List ℕ) : Prop := ∀ (i : ℕ) (h : i < w.length), w[i] ≠ i + 1
instance : DecidablePred IsDerangement := fun w => by unfold IsDerangement; infer_instance
def des (w : List ℕ) : ℕ := (w.zip w.tail).countP fun p => p.2 < p.1
def ders (n k : ℕ) : List (List ℕ) :=
  (List.range' 1 n).permutations.filter fun w => decide (IsDerangement w) && des w == k
def avgFirstLetter (E : List (List ℕ)) : ℚ :=
  (((E.map fun w => w.headD 0).sum : ℕ) : ℚ) / ((E.length : ℕ) : ℚ)
def avgFirst (n k : ℕ) : ℚ := avgFirstLetter (ders n k)
def T (n k : ℕ) : ℕ := (avgFirst n k).den
def rowLen (n : ℕ) : ℕ := if n % 2 = 0 then n - 1 else n - 2
def row (n : ℕ) : List ℕ := (List.range (rowLen n)).map fun j => T n (j + 1)
def statement : Prop := ∀ n : ℕ, 2 ≤ n → T n 1 = 2 ^ (n - 2)
end Math0.Statements.A332010
```

Words are lists of natural numbers; `w[i]` is the letter at the 0-based index i, i.e. at position $i+1$, so `IsDerangement` is the derangement condition. `List.permutations` lists each rearrangement of $[1,\dots,n]$ (`List.range' 1 n`) once, `des` counts the adjacent pairs (w_i, w_{i+1}) with $w_{i+1} < w_i$, and `w.headD 0` is the first letter w_1. Thus `ders n k` lists $\mathcal{D}_{n,k}$ without repetitions, `avgFirst n k` is the average first letter (with the value $0/0 = 0$ if $\mathcal{D}_{n,k} = \emptyset$), and `T n k` is its denominator. Mathlib's rationals are always stored in lowest terms, so `Rat.den` is the denominator of the reduced fraction. For $n \geq 2$ the natural-number subtraction $n-2$ is exact, and $\mathcal{D}_{n,1}$ is nonempty (it contains $23\cdots n1$), so $T(n,1)$ is never the value 1 coming from $0/0$.

The same file proves that `ders n k` lists exactly the derangements of $[n]$ with k descents (`mem_ders`: `w ∈ ders n k ↔ w.Perm (List.range' 1 n) ∧ IsDerangement w ∧ des w = k`), each once (`nodup_ders`), and it checks the definitions against the data of the entry:

```
theorem a_data : (List.range 7).flatMap (fun i => row (i + 2)) =
    [1, 2, 4, 1, 1, 8, 3, 4, 16, 104, 40, 24, 1, 32, 392, 896, 480, 54, 64, 152, 308, 1496, 63,
      108, 1] := by
  simp only [row_eq_rowC]
  decide +kernel
```

Since `List.permutations` uses well-founded recursion, which the kernel does not evaluate efficiently, the check goes through a copy `rowC` of `row` that enumerates words with the structurally recursive `List.permutations'` and uses recursive tests `derB` and `desB`, proved equal to `IsDerangement` and `des` (`derB_one`, `desB_eq`); its word lists are permutations of `ders n k` (`ders_perm`), which changes neither count nor sum (`avgFirstLetter_perm`), whence `row_eq_rowC`. Thus the kernel evaluates rows 2 to 8 (row 8 runs through all $8! = 40320$ words) and matches the first 25 listed terms, including $T(n,1) = 2^{n-2}$ for $2 \leq n \leq 8$; rows 9 to 11 are out of its reach. The theorem `example_T41` checks the worked example of the entry ($\mathcal{D}_{4,1}$, the average $11/4$, and $T(4,1) = 4$).

3. Proof

We follow the Lean file `lean/Math0/Frontier/OEIS/A332010.lean`; each step names the corresponding declaration. Words are finite sequences of natural numbers, $|u|$ is the length of u, juxtaposition is concatenation, and *increasing* means strictly increasing. Positions are numbered from 1.

Definition 3. Let $p \geq 0$. A word $u = u_1 \cdots u_\ell$ is *free at* p if $u_j \neq p + j - 1$ for $1 \leq j \leq \ell$; that is, placed at positions $p, p+1, \dots, p+\ell-1$, no letter of u is at its own position.

In Lean this is `derB u p = true`, where `derB [] p = true` and `derB (x :: xs) p = (x != p && derB xs (p + 1))`. A word is a derangement if and only if it is free at 1 (`derB_one`), and uv is free at p if and only if u is free at p and v is free at $p + |u|$ (`derB_append`).

Lemma 4 (`derB_of_lower`, `lower_of_derB`)**.** *Let u be an increasing word and $p \geq 0$. (1) If every letter of u is $> p$, then u is free at p. (2) If every letter of u is $\geq p$ and u is free at p, then every letter of u is $> p$.*

Proof. (1) Since u is an increasing sequence of integers, $u_j \geq u_1 + j - 1 > p + j - 1$. (2) We have $u_1 \geq p$ and $u_1 \neq p$, so $u_1 > p$, and $u_j \geq u_1$ for all j. In Lean both parts are proved by induction along u, passing from p to $p+1$ for the tail. □

Lemma 5 (`derB_of_upper`, `upper_of_derB`)**.** *Let $u = u_1 \cdots u_\ell$ be an increasing word, $p \geq 0$, and $q = p + \ell - 1$ the last position of u when placed at p. (1) If every letter of u is $< q$, then u is free at p. (2) If every letter of u is $\leq q$ and u is free at p, then every letter of u is $< q$.*

Proof. (1) Since u is increasing, $u_j + (\ell - j) \le u_\ell < q = p + \ell - 1$, so $u_j < p + j - 1$. (Lean: `head_add_length_lt` states $u_1 + (\ell - 1) < B$ when all letters are $< B$, and `derB_of_upper` applies it along the list.) (2) We may assume $\ell \ge 1$. The last letter u_ℓ sits at position q, so $u_\ell \ne q$, and with $u_\ell \le q$ this gives $u_\ell < q$; all other letters are smaller than u_ℓ. In Lean this is an induction from the front of the list. □

Lemma 6 (`desB_of_pairwise`, `pairwise_of_desB`, `desB_split`, `split_of_desB`)**.** (1) *An increasing word has no descent, and a word with distinct letters and no descent is increasing.* (2) *For words l, m and letters a, b,* $\mathrm{des}(l\,a\,b\,m) = \mathrm{des}(l\,a) + [\,b < a\,] + \mathrm{des}(b\,m)$*, where $[\,b < a\,]$ is 1 if $b < a$ and 0 otherwise.* (3) *A word w with distinct letters and $\mathrm{des}(w) = 1$ can be written $w = u\,a\,b\,v$ with $u\,a$ and $b\,v$ increasing.*

Proof. (1) and (2) are immediate from the definition (in Lean, inductions on the word using $\mathrm{des}(x\,y\,l) = [\,y < x\,] + \mathrm{des}(y\,l)$). For (3) we argue by induction on $w = x\,y\,l$ (a word with a descent has at least two letters). If $y < x$, the step (x, y) is the only descent, so $y\,l$ has no descent and is increasing by (1); take u empty, $a = x$, $b = y$, $v = l$. Otherwise $x < y$ (the letters are distinct), $\mathrm{des}(y\,l) = 1$, and by induction $y\,l = u'\,a\,b\,v$ with $u'a$ and bv increasing. Every letter of $u'a$ is $\ge y > x$, so $x\,u'\,a$ is increasing, and $w = (x\,u')\,a\,b\,v$. □

Fix $m \ge 0$ and let $R_m = 2\,3 \cdots (m+1)$ be the increasing word on $\{2, \dots, m+1\}$ (`List.range' 2 m`). The subwords (sublists) of R_m are exactly the increasing words with letters in $\{2, \dots, m+1\}$, one for each subset; Mathlib's `List.sublists'` lists the 2^m of them, each once.

Definition 7 (`word`, `words`)**.** For a sublist s of R_m let $\bar{s}$ be the sublist of R_m formed by the letters not in s, and

$$W_m(s) = s\,(m+2)\,1\,\bar{s}.$$

Let $\mathcal{W}_m$ be the list of the words $W_m(s)$, with s running over the sublists of R_m.

For $m = 2$ the sublists $\emptyset, 2, 3, 23$ give 4123, 2413, 3412, 2341, the four words of the example in A332010.

Proposition 8 (`word_mem_ders`)**.** *For every sublist s of R_m, $W_m(s)$ is a derangement of $[m+2]$ with exactly one descent.*

Proof. Write $v = \bar{s}$, so $W = W_m(s) = s\,(m+2)\,1\,v$. The words s and v are increasing, their letters lie in $\{2, \dots, m+1\}$, and each element of $\{2, \dots, m+1\}$ is a letter of exactly one of them. Hence W has distinct letters and its set of letters is $[m+2]$, so W is a rearrangement of $1\,2 \cdots (m+2)$ (`List.perm_ext_iff_of_nodup`); comparing lengths, $|s| + |v| = m$.

No fixed point. By `derB_append`, W is free at 1 if and only if: s is free at 1, which holds by Lemma 4(1) since its letters are ≥ 2; the letter $m+2$ is not at position $|s| + 1 \le m + 1$; the letter 1 is not at position $|s| + 2 \ge 2$; and v is free at $|s| + 3$. For the last condition, v occupies positions $|s| + 3, \dots, |s| + 2 + |v| = m + 2$ and its letters are $\le m + 1 < m + 2$, so Lemma 5(1) applies.

One descent. The words $s\,(m+2)$ and $1\,v$ are increasing, because the letters of s are $\le m+1$ and those of v are ≥ 2. By Lemma 6(1),(2), $\mathrm{des}(W) = 0 + [\,1 < m+2\,] + 0 = 1$. □

Proposition 9 (`mem_words_of_mem_ders`)**.** *Every derangement w of $[m+2]$ with exactly one descent equals $W_m(s)$ for some sublist s of R_m.*

Proof. By Lemma 6(3), $w = u\,a\,b\,v$ with $P = u\,a$ and $Q = b\,v$ increasing, and $|P| + |Q| = m + 2$. Since w is free at 1, P is free at 1 and Q is free at $|P| + 1$, and the last position of Q is $m + 2$. All letters of w lie in $[m+2]$. Lemma 4(2) shows that every letter of P is > 1, and Lemma 5(2) that every letter of Q is $< m + 2$.

Hence the letter 1 of w lies in Q, and since b is the smallest letter of Q and $b \geq 1$, we get $b = 1$. Likewise the letter $m+2$ lies in P, and since a is the largest letter of P and $a \leq m+2$, we get $a = m+2$. So $w = u\,(m+2)\,1\,v$.

The word u is increasing and its letters lie in $\{2, \ldots, m+1\}$ (they are > 1 and $< a$); an increasing word with distinct letters from R_m is a sublist of R_m (`List.subperm_of_subset`, `List.sublist_of_subperm_of_pairwise`). It remains to show $v = \bar{u}$. Both words are increasing, so it suffices that they have the same letters (`List.Pairwise.eq_of_mem_iff`). A letter x of v satisfies $1 < x < m+2$ (since $Q = 1\,v$ is increasing and its letters are $< m+2$), and x is not a letter of u because w has distinct letters; so x is a letter of $\bar{u}$. Conversely, if $2 \leq x \leq m+1$ and x is not a letter of u, then x is a letter of w different from $m+2$ and 1, so it is a letter of v. Hence $w = W_m(u)$. □

Lemma 10 (`words_nodup`)**.** *The map $s \mapsto W_m(s)$ is injective on the sublists of R_m; hence $\mathcal{W}_m$ has no repetitions.*

Proof. The letter $m+2$ occurs neither in s nor in $1\,\bar{s}$, so s is the part of $W_m(s)$ before the letter $m+2$ (`List.append_cons_inj_of_notMem`). The list of sublists of R_m has no repetitions since R_m has none. □

Corollary 11 (`ders_perm_words`)**.** *The list `ders (m+2) 1` of $\mathcal{D}_{m+2,1}$ is a permutation of $\mathcal{W}_m$. Consequently $|\mathcal{D}_{m+2,1}| = 2^m$ and $\sum_{w \in \mathcal{D}_{m+2,1}} w_1 = \sum_s \big($first letter of $W_m(s)\big)$, the sum over the sublists s of R_m.*

Proof. Both lists have no repetitions (`nodup_ders` and Lemma 10) and they have the same elements (`mem_ders` with Propositions 8 and 9), so one is a permutation of the other (`List.perm_ext_iff_of_nodup`). Length and sum of first letters are invariant under permutation (`avgFirstLetter_perm`). □

For a word s and $c \geq 0$ let $h_c(s)$ be the first letter of s if s is nonempty, and c if s is empty (`s.headD c`). The first letter of $W_m(s)$ is $h_{m+2}(s)$.

Lemma 12 (`sum_heads`)**.** *For $a, k \geq 0$ let $F(a,k) = \sum_s h_{a+k}(s)$, the sum over the 2^k sublists s of $a\,(a+1)\cdots(a+k-1)$. Then $F(a,k) + 1 = (a+1)\,2^k$.*

Proof. Induction on k, for all a simultaneously. For $k = 0$ the only sublist is empty and $F(a,0) = a$. For the step, the sublists of $a\,(a+1)\cdots(a+k)$ are the sublists s of $(a+1)\cdots(a+k)$ together with the words $a\,s$ (`List.sublists'_cons`). The first group contributes $F(a+1,k)$, because $h_{a+(k+1)} = h_{(a+1)+k}$; the second contributes $a \cdot 2^k$. By the induction hypothesis,

$$F(a,k+1) + 1 = \big(F(a+1,k) + 1\big) + a\,2^k = (a+2)\,2^k + a\,2^k = (a+1)\,2^{k+1}. \qquad \square$$

Proof of Theorem 1. Write $n = m+2$. By Corollary 11, $|\mathcal{D}_{n,1}| = 2^m$ and

$$N := \sum_{w \in \mathcal{D}_{n,1}} w_1 = \sum_s h_{m+2}(s) = F(2,m),$$

so $N + 1 = 3 \cdot 2^m$ by Lemma 12. This proves (1) and (2), and the average first letter is $N/2^m$. The integers N and 2^m are coprime: for $m = 0$ trivially, and for $m \geq 1$ because $N = 3 \cdot 2^m - 1$ is odd. Hence the fraction $N/2^m$ is in lowest terms and its denominator is $2^m = 2^{n-2}$ (Mathlib's `Rat.den_div_eq_of_coprime`). Finally $N/2^m = 3 - 2^{-m} = 3 - 2^{2-n}$. □

Corollary 2 follows: by Theorem 1(3) the numerator is $3 \cdot 2^{n-2} - 1 = A083329(n-1)$ [10].

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A332010.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A332010.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A332010.result`, of type `Math0.Statements.A332010.statement`; the auxiliary declarations of this file named in Section 3 are private to it, while `derB_one`, `mem_ders`, `nodup_ders` and `avgFirstLetter_perm` are public lemmas of the statement file. The tactic proof of `result` follows the proof of Theorem 1 step by step (parts (1) and (2) appear as the facts `hlen` and `hsum`). The file's last line is the audit command

```
#math0_settles Math0.Frontier.A332010.result Math0.Statements.A332010.statement
```

Neither file contains `sorry`, `native_decide`, `set_option` or an axiom declaration; the data checks in the statement file use `decide +kernel`, which adds no axioms.

Audit. The command `#math0_settles` (defined in `lean/Math0/Audit.lean`, for statements formalized by math0) fails unless `statement` is a universe-monomorphic `Prop`-valued definition, both `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (which excludes `sorryAx` and `native_decide`), and the type of `result` is syntactically `statement` or its negation, without unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A332010.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A332010.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A332010.result' proves 'Math0.Statements.A332010.statement'
```

Toolchain and reproduction. We use Lean 4 [9] `v4.33.1` and Mathlib [8] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [4] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh`: it fetches the Mathlib cache, builds all math0 results (re-running every audit), and checks that the pinned dependencies are unmodified. This result alone is checked in `lean/` by `lake build Math0.Statements.OEIS.A332010` (about a minute of CPU time, mostly `a_data`) and `lake env lean Math0/Frontier/OEIS/A332010.lean`.

References

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Produced by Claude (an AI model developed by Anthropic) in the math0 project